
EE1060: Differential Equations and Transform Techniques

Lecture 34: Laplace Transform and Differential Equations

Topics :

1. Inverse Laplace Transform
 2. Convolution
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Inverse Laplace Transform - Partial Fractions

given $H(s) = \frac{N(s)}{D(s)}$ Degree of $N(s)$ (m) < Degree of $D(s)$ (n)

linear factors: $D(s) = (s - p_1) \cdots (s - p_n)$ where $p_1, \dots, p_n$ are poles & distinct.

$$H(s) = \sum_{i=1}^n \frac{A_i}{(s - p_i)}$$

$$h(t) = \mathcal{L}^{-1}\{H(s)\} = \sum A_i e^{p_i t} u(t)$$

if $p_i < 0$, $\Rightarrow$ Poles are on left half of s-plane

$h(t) \rightarrow 0$ as $t \rightarrow \infty$, stable.

Repeated linear factors: $D(s) = (s - p_1) \cdots (s - p_{k-1}) (s - p_k)^{n-k}$

$$H(s) = \sum_{i=1}^{k-1} \frac{A_i}{(s - p_i)} + \frac{A_k}{s - p_k} + \frac{A_{k+1}}{(s - p_k)^2} + \cdots$$

$$h(t) = \left[\sum_{i=1}^{k-1} A_i e^{p_i t} + A_k e^{p_k t} + A_{k+1} t e^{p_k t} + A_{k+2} \frac{t^2}{2} e^{p_k t} + \cdots \right] u(t)$$

Inverse Laplace Transform - Partial Fractions

Irreducible Quadratic form:- $D(s) = (s^2 + as + b) \cdots$ where it cannot be written as $(s+a)^2$

Corresponding Sample term $\rightarrow$

$$= \frac{As+B}{s^2+as+b} + \dots$$

Repeated Quadratic: $D(s) = (s^2 + as + b)^2 (\quad) \dots$

$$= \frac{A_1s+B_1}{s^2+as+b} + \frac{A_2s+B_2}{(s^2+as+b)^2} + \dots$$

Example:- $H(s) = \frac{1}{s^2 + 2\xi\omega_n s + \omega_n^2}$ $a = 2\xi\omega_n$ and $b = \omega_n^2$

$$= \frac{1}{s^2 + 2\xi\omega_n s + \omega_n^2 \xi^2 + \underbrace{(1-\xi^2)\omega_n^2}_{\omega_d^2}}$$

$\omega_d = \omega_n \sqrt{1-\xi^2}$

$$= \frac{1}{s^2 + 2\xi\omega_n s + \omega_n^2 \xi^2 + \omega_d^2} = \frac{1}{(s + \xi\omega_n)^2 + \omega_d^2}$$

$\xi = 0 \Rightarrow H(s) = \frac{1}{s^2 + \omega_n^2} \Rightarrow h(t) = \frac{1}{\omega_n} \sin(\omega_n t) u(t)$

Examples

 $|\xi| < 1 :$

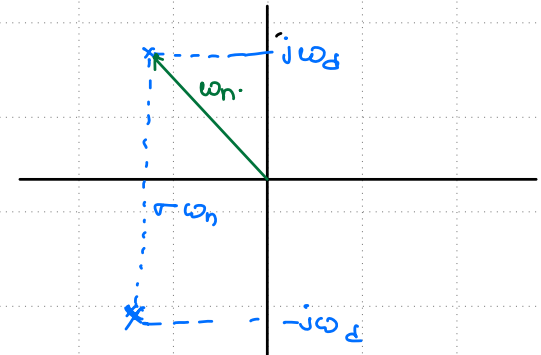
$$H(s) = \frac{1}{(s + \xi\omega_n)^2 + \omega_d^2}$$

$$\omega_d = \omega_n \sqrt{1 - \xi^2}$$

$$\text{Poles} \rightarrow -\xi\omega_n \pm j\omega_d$$

$$\Rightarrow h(t) = \frac{1}{\omega_d} e^{-\xi\omega_n t} \sin(\omega_d t)$$

$0 < \xi < 1 :$ Poles are on the left half of s-plane.
 $h(t) \rightarrow 0$ as $t \rightarrow \infty$



$-1 < \xi < 0 :$ Poles are on the Right half of s-plane
 $h(t) \rightarrow \infty$ as $t \rightarrow \infty$

 $\xi = 1 :$

$$H(s) = \frac{1}{(s + \omega_n)^2}$$

$$h(t) = e^{-\omega_n t} \underbrace{t u(t)} \rightarrow 0 \text{ as } t \rightarrow \infty; \text{ Stable.}$$

 $|\xi| > 1 :-$

Practice problem.

Examples

Example 1 :- $H(s) = \frac{s}{s^2 + 2\xi\omega_n s + \omega_n^2}$ $|\xi| < 1$:

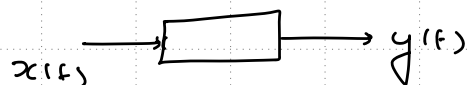
$$= \frac{s + \xi\omega_n - \xi\omega_n}{(s + \xi\omega_n)^2 + \omega_d^2} \Rightarrow \frac{s + \xi\omega_n}{(s + \xi\omega_n)^2 + \omega_d^2} - \frac{\xi\omega_n}{(s + \xi\omega_n)^2 + \omega_d^2}$$

$$= e^{-\xi\omega_n t} \left[\cos \omega_d t - \frac{\xi\omega_n}{\omega_d} \sin \omega_d t \right]$$

Example 2 :- $H(s) = \frac{1}{s(s^2 + 2\xi\omega_n s + \omega_n^2)}$ $|\xi| < 1$

$h(t) \rightarrow ?$

Convolution and System Response



a) find $h(t) \rightarrow ?$

b) compute $y(t) = x(t) * h(t)$

using the Laplace Transform:-

$$\begin{aligned}
 Y(s) &= \int_{-\infty}^{\infty} [x(t) * h(t)] e^{-st} dt \\
 &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau \right] e^{-st} dt \\
 &= \int_{-\infty}^{\infty} x(\tau) \left(\int_{-\infty}^{\infty} h(t-\tau) e^{-st} dt \right) d\tau \\
 &= \int_{-\infty}^{\infty} x(\tau) \left(\int_{-\infty}^{\infty} h(t_1) e^{-st_1} e^{-s\tau} dt_1 \right) d\tau
 \end{aligned}$$

$$Y(s) = X(s) H(s)$$

$$\Rightarrow H(s) = \frac{Y(s)}{X(s)} \leftarrow \text{Transfer function.}$$